



Leading Digital Trust

Cyberwar, Digital Public Infrastructure (DPI), and the New Frontline of National Security

A Position Paper Defining a New Global Risk Category and the Strategic Responses Required to Strengthen DPI Security in an Era of Persistent, Cyber-Assisted Conflict

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ABOUT THE POSITION PAPER

Modern warfare has no front line. It has no uniform, no declaration, and often, no explosion. What it has is intent, persistence, and patience. Today's adversaries do not merely attack systems; they inhabit them, mapping vulnerabilities across months or years before a single command is executed.

The battlefield is invisible; however, the consequences are not.

This position paper examines the convergence of two defining dynamics of the current era: the escalating sophistication of state and non-state cyberwarfare, and the rapid global expansion of Digital Public Infrastructure (DPI). As nations in the Global South accelerate deployment of digital identity, payment rails, and data exchange frameworks, they are simultaneously enlarging the attack surface available to adversaries, many of whom operate with strategic patience and technological parity.

The paper is structured around four interlocking arguments:

- 1) Cyberwar is a persistent condition, not a discrete event, and its defining characteristic is positioning, not disruption.
- 2) AI has become a force multiplier for attackers, compressing timelines, scaling social engineering, and enabling autonomous intrusion at speeds that outpace human response.
- 3) DPI systems, once considered tools of development, have crossed the threshold into critical national infrastructure and must be governed and secured accordingly.
- 4) The pathway forward requires integrating security-by-design into DPI governance, aided by Independent validations, strengthening independent assurance mechanisms, and building multilateral cyber-resilience frameworks calibrated for Global South realities.

“The absence of visible failure cannot define security, but by the system’s ability to withstand and operate under active threat conditions.”

Key narratives that are synthesized in this position paper:

The arguments that follow outline the structural realities redefining Digital Public Infrastructure in the current cyberwar era and the critical gaps that must be addressed to secure its next phase of deployment and scale (decade: 2026 – 2036).

- a) Cyberwar is persistent, not episodic
→ DPI operates inside an active threat environment
- b) DPI is now a critical national infrastructure
→ Not development tech anymore — national security asset
- c) Security → Safety → Trust is a dependency chain
→ Misunderstood globally
- d) The real gap is NOT technology — it’s governance + independence
→ No independent DPI security assurance layer exists
- e) Systemic risk = DPG reuse + ecosystem blindness
→ Vulnerabilities propagate across countries

Taken together, these dynamics signal a fundamental shift: securing DPI is no longer a technical exercise, but a strategic imperative requiring new governance models, independent assurance, and ecosystem-level coordination.

EXECUTIVE SUMMARY

Modern cyber conflict is not an event-driven phenomenon. It is a persistent condition. Adversaries no longer wait for moments of crisis to strike; they operate continuously, embedding themselves within systems, mapping dependencies, and preparing to activate at strategically advantageous times. In this environment, Digital Public Infrastructure (DPI) does not operate in peacetime conditions. It operates within an active, contested threat landscape, where exposure is constant, and the invisibility of an attack does not equate to the absence of compromise.

At the same time, DPI has undergone a structural reclassification. What began as a tool for development, enabling financial inclusion, digital identity, and service delivery, has now crossed a critical threshold. DPI systems underpin identity, payments, and data exchange at the population scale. Their disruption affects not users, but citizens, not transactions, but economies. As such, DPI must now be understood and governed as critical national infrastructure, with direct implications for sovereignty, stability, and national security.

Yet the conceptual foundations guiding DPI design and investment remain misaligned with this reality. Security, safety, and trust are frequently treated as parallel priorities, to be balanced or optimized independently. This is a structural error. These three dimensions form a dependency chain: security is the prerequisite, safety is the lived experience, and trust is the outcome. When security fails, safety mechanisms collapse, and trust erodes, often irreversibly. This misunderstanding persists, producing systems that are technically functional but structurally fragile.

The central challenge, however, is not technological. The tools, standards, and controls required to secure DPI systems largely exist. The failure lies elsewhere. It lies in governance, in institutional design, and in the absence of independence. Today, most DPI security assessments are conducted by the same actors who design, build, or operate the systems being evaluated, creating a structural conflict of interest and leaving no entity positioned to identify systemic risk across platforms, countries, and shared components. There is, at present, no independent, ecosystem-level DPI cybersecurity assurance capability, despite the scale and criticality of the systems involved.

This gap is amplified by DPI's defining feature: reuse. The Digital Public Goods (DPG) model enables rapid deployment and scalability through shared open-source components. But this same model introduces a multiplier effect for risk. A vulnerability in a widely deployed component is not contained within a single system or country. It propagates silently across every implementation that depends on it. In the absence of coordinated visibility and cross-border assurance mechanisms, this creates a condition of systemic exposure combined with ecosystem blindness, in which risks are replicated faster than they are detected.

A critical asymmetry defines the current landscape: countries in the Global South are leading the rapid adoption and scaling of Digital Public Infrastructure, yet they do so with significantly lower levels of national cybersecurity preparedness, institutional capacity, and security investment. This creates a widening exposure gap, in which the most digitally transformative systems are being deployed in the least-defended environments. In contrast, countries in the Global North, while advancing more cautiously in DPI adoption, have invested heavily in national cybersecurity and defense capabilities, often allocating resources orders of magnitude greater (hundreds of times more in some of the advanced economies) than those directed toward digital government services, effectively reversing the risk profile between adoption and protection.

Taken together, these dynamics define the central focal point of this paper:

Digital Public Infrastructure now operates at the intersection of cyberwarfare, national security, and global digital development. Still, it lacks the governance models, assurance mechanisms, or conceptual frameworks needed to secure it, at a minimum. The sections that follow examine this gap in detail and outline the structural, technical, and institutional responses required to close it.

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I. Introduction: A Quiet Battlefield Redefining Power

Something shifted like conflict, not with a single incident, but gradually, almost imperceptibly. Armies still exist, and missiles are still launched. But the decisive terrain is no longer found solely in physical geography. It lives in submarine cables, cloud environments, payment systems, biometric databases, and the digital rails that carry trust instead of goods.

Modern conflict has undergone a profound structural transformation.

It is no longer defined solely by physical confrontation or territorial control; instead, it increasingly unfolds within the digital systems that underpin societies: financial networks, identity platforms, payment rails, and data exchange infrastructures. These systems, once considered tools of efficiency and development, are now deeply embedded in national security calculations.

Cyberwarfare reflects this transformation. It is not characterized by visible destruction, but by persistence, subtlety, and strategic positioning. It operates through infiltration rather than confrontation, through access rather than immediate disruption.

A recent regional risk snapshot from Thomas Murray, covering GCC member states in April 2026, illustrates the paradox at the heart of modern cyber conflict with striking precision:

“Markets are operating as normal. Financial institutions, including the stock exchanges, central securities depositories, and central banks, are currently operating normally across Bahrain, Egypt, Jordan, Kuwait, Lebanon, Oman, Qatar, Saudi Arabia, and the UAE. During the past week, a total of 38 cyber incidents were recorded across GCC member states, representing a 38.7% decrease from the previous week’s 62 incidents.” — **Thomas Murray, Regional Risk Update, April 2026**

Read that again. Thirty-eight cyber incidents in a single week across a region described as “operating as normal.” This is the paradox. Systems appear stable, yet threats remain constant and evolving beneath the surface. Stability and security are not the same thing. In modern cyberwar, one is often performed while the other is actively eroded.

Website defacement, the digital equivalent of spray-painting a building, dominated the visible attack category, consistent with ideologically motivated hacktivist campaigns. But beneath that visible layer, data leaks and credential-harvesting operations continued in parallel: quieter, slower, and far more consequential in their long-term implications.

This two-tier architecture of modern cyberattacks, visible noise above, strategic positioning below, is not accidental. It is doctrine.

Given the above, the implication is immediate and consequential. If the terrain of conflict has shifted into the digital systems that underpin economies and governance, then the security of those systems is no longer a technical concern but a matter of national resilience and strategic stability. Systems that appear operational can no longer be assumed to be secure, and policies built on that assumption are fundamentally flawed.

The ability of a state to function, deliver services, and maintain public trust now depends on its capacity to detect, withstand, and operate under continuous, often invisible, cyber pressure. In this context, failing to treat Digital Public Infrastructure as contested terrain is not a gap in awareness; it is a direct exposure to systemic and potentially irreversible disruption.

II. The Architecture of Modern Cyberwar

2.1 Persistent, Strategic, and Patient

Cyberwarfare today is best understood as a continuous strategic engagement, rather than a series of isolated incidents. The presence of dozens of cyber incidents within a single week, even in relatively stable geopolitical conditions, demonstrates that digital systems are under constant observation, probing, and pressure.

The logic is as old as warfare itself, applied with new precision: control the terrain before you need to fight over it. Control the system before you need to disrupt it.

At the surface, visible attacks serve symbolic and communicative purposes, signaling presence, demonstrating capability, and shaping public perception. Beneath this layer lies a more consequential set of activities: establishing persistent access, mapping system architectures and dependencies, and accumulating credentials for future use. This distinction is critical.

Cyberwar is not primarily about immediate disruption. It is about positioning.

This strategic patience distinguishes cyberwar from traditional forms of conflict. Adversaries invest time and resources to gain access that may be exploited only at times of heightened political or economic sensitivity, such as elections, diplomatic negotiations, financial crises, or periods of social unrest.

The fact that there are no visible events today in your backyard does not mean you're not sized for later exploitation when the time and needs silently enter, usually long before you know it!

2.2 AI As The Attack Force Multiplier That Changed Everything

If cyberwarfare has always been strategic, artificial intelligence has made it fast. Brutally, asymmetrically fast.

The CrowdStrike 2026 Global Threat Report documents the scale of this shift with arresting clarity:

Indicator	Metric
Fastest recorded eCrime breakout time	27 seconds
Increase in attacks by AI-enabled adversaries (2025)	+89%
Malware-free detections in 2025	82%
Organizations with AI tools weaponized against them	90+
ChatGPT mentions in criminal forums vs. any other model	+550%

Eighty-two percent of detections in 2025 involved no traditional malware whatsoever. Attackers leveraged legitimate tools and living-off-the-land techniques, using the target's own systems against it. This is not a technical refinement. It is a strategic philosophy change.

AI now operates as a dual threat. It accelerates offense while simultaneously expanding the attack surface. In 2025, more than 90 organizations had legitimate AI tools exploited to generate malicious commands or exfiltrate sensitive data. Meanwhile, both OpenAI and Google have confirmed identifying usage from state-sponsored actors attempting to augment influence operations, phishing campaigns, and malicious code generation through large language models.

“2025 could see the first major breach caused entirely by an autonomous AI agent operating within a target’s network — self-propagating, adaptive, and making decisions without direct hacker oversight.” — DeepStrike Cybersecurity Research, 2025

We are not waiting for AI-driven attacks. We are already living through them. The question is whether defenders can adapt as rapidly as adversaries are already doing.

What we are seeing in the attacker landscape today strongly implies AI has become a force multiplier not just in scale, but in **speed, precision, and asymmetry**, fundamentally altering the balance between attacker and defender. What once required coordinated human effort, time, and expertise can now be executed autonomously, iteratively, and at machine speed. The result is not simply more attacks—it is a different class of attack altogether, in which reconnaissance, exploitation, and adaptation occur in near real time. This shift compresses the defender’s response window from days or hours to seconds, eroding the effectiveness of traditional detection and response models.

- Attack timelines have collapsed from hours to seconds, reducing the viability of human intervention
- Social engineering has scaled exponentially through AI-generated, context-aware interactions
- Living-off-the-land techniques now blend seamlessly with legitimate system behavior
- AI enables continuous probing and adaptive attack paths without human oversight

This is not an evolution of cybercrime; it is a **redefinition of operational doctrine**.

AI now poses a dual-layer threat: it enhances adversarial capabilities while simultaneously expanding the attack surface through its integration into systems. As AI becomes embedded in identity verification, fraud detection, citizen interfaces, and decision-making processes within DPI, it introduces new vulnerabilities that are not yet fully understood, governed, or tested at scale. Attackers are no longer limited to exploiting systems; they are increasingly exploiting the intelligence layer within those systems.

- AI systems introduce new attack vectors: model inversion, data poisoning, prompt injection
- Autonomous agents can persist, adapt, and evade detection over extended periods
- AI-generated code lowers the barrier to sophisticated attack execution
- Integration of AI into critical systems expands the blast radius of compromise
- Governance and oversight mechanisms for AI remain immature or absent in most deployments

The result is a compounding effect: more powerful attackers targeting more complex, less well-governed systems.

The strategic implication is unavoidable: AI has shifted cyberwar from a contest of capability to a contest of adaptation speed, where advantage accrues to those who can learn, iterate, and act faster than their opponent. In this environment, static defenses, periodic assessments, and human-paced response models are structurally misaligned with the threat. The challenge is no longer whether attacks can be prevented, but whether systems can remain resilient under continuous, intelligent, and autonomous pressure.

“AI has not just accelerated cyberwar; it has changed its nature. And in doing so, it has raised the cost of inaction to a level that most systems are not yet designed to withstand.”

2.3 The Illusion of Stability

One of the most persistent misconceptions in cyber risk assessment is this: operational continuity equates to security. It does not.

Modern cyber operations are often explicitly designed to preserve the appearance of normalcy. Adversaries avoid triggering defensive responses, maintain long-term access, and operate within acceptable thresholds of disruption. A system that continues to function may still be under surveillance, partially compromised, or vulnerable to the activation of pre-positioned access points at the adversary's choosing.

This requires a fundamental shift in how policymakers and system operators understand security:

1. **From reactive validation:** “nothing has failed; therefore, we are safe.”
2. **To proactive and continuous assurance:** “We understand and can verify the system's security posture under active threat conditions and evolving threat landscape.”

The GCC snapshot is instructive again: three individuals accounted for over 63% of all cyber incidents across the region in a single week.

The current threat landscape is shaped disproportionately by a handful of persistent, high-volume operators. This concentration should unsettle any assumption that broad surveillance or bulk defense is sufficient.

The implication is clear: systems judged secure based on visible stability rather than verified resilience operate under a false sense of assurance, leaving critical infrastructure exposed to controlled, delayed, and potentially systemic disruption.

2.4 Attrition, Not Confrontation

Cyberwarfare does not seek decisive, immediate outcomes. Instead, it exerts gradual, cumulative pressure on systems and institutions. Each incident, no matter how minor it may appear, contributes to expanding the attack surface, revealing system behaviors and weaknesses, and incrementally weakening institutional resilience.

This model resembles **attrition rather than confrontation**. It operates below the threshold of conventional war, making it difficult to attribute, regulate, or respond to through existing legal and diplomatic frameworks.

Over time, this accumulation leads to increased vulnerability to large-scale disruption, erosion of public trust, and strategic asymmetry between attacker and defender. The consequences of cyberwar are not always immediate; they often emerge gradually, as vulnerabilities accumulate and are eventually exploited.

Cyberwar is not episodic (a series of events), but a persistent operating condition, and systems not designed, governed, and continuously validated to withstand active threats will predictably fail at scale, on terms set by the attacker, with consequences including economic instability, loss of trust, and national vulnerability.

“In cyberwar, the question is no longer if systems will be tested or not but whether they will hold when they are.”

III. The Global Threat Landscape: Evidence, Actors, and Economics

3.1 State Actors and the Proxy Doctrine

The attribution challenge is one of cyberwar's most powerful strategic assets, not a weakness of the defender, but a deliberate design of the attacker. Governments routinely leverage proxy groups to conduct offensive operations, enabling plausible deniability while maintaining strategic influence. The line between state and non-state actors has become deliberately, strategically blurred.

Reported numbers that have emerged from 2025 intelligence assessments paint a clear picture of the dominant actors:

Indicator	Metric
China: share of state-backed campaigns against governments (mid-2025)	28%
Russia: share of offensive cyber actions (2025)	22%
Russia's primary target region (Jul 2023 – Jun 2024)	Middle East & N. Africa (52%)
Countries targeted by Iranian infrastructure intrusions (2025)	Latin America (8 incidents)
North Korea crypto exchange raids (early 2025)	>\$105 million
Multinational cyber offensives recorded, Q1 2025	17
False-flag masking tactics across operations	15%
Countries where election meddling was attempted	18

More reports suggest that Germany offers a stark case study in the real-world cost of state-linked cyber operations. In 2025, cyberattacks linked to perpetrators aligned with foreign intelligence agencies were estimated to have wiped approximately €300 billion from the German economy. This extraordinary figure underscores the systemic economic risk posed by geopolitically motivated cyber activity.

Media reports say Russia's Qilin ransomware group claimed responsibility in March 2026 for an attack on the German democratic socialist party Die Linke, explicitly framing it as "hybrid warfare" linked to Moscow's broader geopolitical goals. China-linked group UNC3886 breached all four of Singapore's major telecommunications carriers. Handala, an Iranian-linked group, simultaneously triggered factor resets across 200,000 corporate devices in 79 countries during an attack on medical device maker Stryker.

"These are not isolated incidents. They are symptoms of a systemic condition."

NB. The observations and metrics in sections 3.1 and 3.2 are based solely on reported intelligence and publicly available research and are included for contextual analysis only, without constituting independent verification or attribution by the author or Nucleus Systems.

3.2 The Economics of Cyberwar: Numbers That Should Alarm

The financial data is now unambiguous, and it demands attention at the highest levels of governments

Indicator	Metric
Global cybercrime losses forecast (2026)	>\$10.5 trillion
FBI reported cybercrime losses, U.S. alone (2025)	\$20.9 billion (+26% YoY)
IC3 complaints received in 2025	>1 million (first time)
Global cybersecurity spending (2026 est.)	\$240 billion
Cyberwarfare market size (2025)	\$211.6 billion
Average ransomware payout vs. annual cyber budget (2026)	Exceeds budget in 52% of orgs
Average cost of a state-linked cyberattack	\$3.6 million
Ransomware increase, Q1 2024 to Q1 2025	+169% (572 to 1,537 incidents)
Healthcare breach cost per incident (2025)	\$7.42 million avg

“If cybercrime were a country, it would be the third-largest economy in the world, behind only the United States and China.” — Cybersecurity Ventures, 2026 Projection.

Ransomware alone has become a weapon of strategic disruption: involved in 44% of all data breaches in 2025, with active groups surging globally and the United States remaining the most targeted nation. But the economic carnage is matched by the human cost, in healthcare, where breaches now average \$7.42 million per incident, 28% of surveyed providers reported increased patient mortality rates attributable to cyberattack-induced care disruptions. **“These are not just statistics. They are policy failures waiting to be named”.**

3.3 Critical Infrastructure: The Strategic Target

Critical infrastructure has become the center of gravity for state-sponsored cyber operations. The CSIS Significant Cyber Incidents tracker for early 2026 alone documents attacks on:

1. Poland’s energy grid (Russia-linked Electrum, January 2026)
2. Singapore’s four major telecommunications carriers (China-linked UNC3886, February 2026)
3. The European Commission’s cloud infrastructure (March 2026)
4. Medical device maker Stryker operates across 79 countries (Iranian-linked Handala, March 2026)
5. Canada’s Telus — 700TB of data exfiltrated by ShinyHunters (March 2026)
6. Germany’s Die Linke political party systems (Russia-linked Qilin, March 2026)

Supply chain vulnerabilities represent a particularly acute dimension of this threat. In 2025, attackers inserted malicious code into widely used open-source libraries through a technique known as dependency hijacking or repo jacking, affecting thousands of downstream projects simultaneously.

For DPI ecosystems built on open-source components, this is not a hypothetical risk. It is an active threat vector. By 2026, security analysts project that large enterprises may begin requiring all critical vendors to provide Software Bills of Materials (SBOMs) and to submit to continuous monitoring. This supply chain discipline must apply equally to DPI component ecosystems.

IV. Digital Public Infrastructure: Status, Risk, and Pathway

A THREE-PART FRAMEWORK FOR THE CRITICAL DECADE AHEAD

Digital Public Infrastructure is the defining development story of the early twenty-first century. It is also increasingly becoming a national security story. The two cannot be separated.

The three sections that follow represent a consolidated analytical framework, replacing a more fragmented treatment of DPI themes. They move from the descriptive (what is the status quo) through the diagnostic (what are the risks and their implications) to the prescriptive (what are the viable pathways forward). Together, they form a coherent strategic argument: that DPI's greatest asset, its scale, is also its greatest liability without a commensurate security architecture.

4.1 The Current Status Quo: DPI at Scale

The Numbers Are Staggering Globally

As of 2026, according to <https://dpimap.org/>, at least 70 countries operate DPI-like digital identity systems, 102 countries have DPI-like digital payment systems, and 103 countries have DPI-like data exchange systems. This is not gradual evolution. It is a structural transformation of the global public sector at a pace that has few historical precedents.

India's DPI serves as both the benchmark and the blueprint. The Aadhaar biometric identity system has enrolled over 1.3 billion residents. The Unified Payments Interface (UPI) broke through the 20 billion monthly transaction thresholds in August 2025. The cost of identity verification fell from \$10–20 per transaction under the old system to \$0.27, a 98% reduction that has enabled millions to access government benefits and banking services directly.

“India’s DPI demonstrates what is possible when you treat digital identity, payments, and data exchange as secure foundational infrastructure rather than a profit center.” — World Economic Forum, October 2025

The “50-in-5” coalition, which seeks to help 50 countries establish DPI within 5 years, drawing heavily on India's model, signals global ambition. Brazil has led DPI integration across Latin America and served as a platform for South-South knowledge exchange during its G20 presidency. Six African nations (Sierra Leone, Tanzania, Kenya, Ethiopia, The Gambia, and Lesotho) have now formally adopted elements of India Stack through bilateral agreements signed in February 2026.

Tanzania exemplifies the pace of this expansion. By mid-2025, more than 25 million Tanzanians had been issued a Jamii Number, with the country's Jamii X-Change supporting data sharing and Jamii Pay enabling digital transactions. These are not pilot programs. They are national systems.

The DPG Ecosystem: Strength Through Openness and Reuse

Underpinning much of this expansion is the Digital Public Goods (DPG) ecosystem: a set of reusables. These open-source components have enabled rapid deployment across multiple countries and interoperability between systems. The Modular Open Source Identity Platform (MOSIP), developed in Bangalore, is now deployed in more than twenty countries. Ukraine's Diia platform is being made available for export as open-source code, with USAID pledging at least \$650,000 to support its proliferation across South America and sub-Saharan Africa.

This model has delivered extraordinary results, efficiency through reuse. Accelerated innovation. Broad accessibility. Reduced the cost of entry for countries with limited institutional capacity.

DPI is no longer a distant vision. It is already shaping how billions live and interact every day. That is precisely what makes the security conversation so urgent.

South Africa: A Case Worth Watching

President Cyril Ramaphosa, in his Opening of Parliament Address, explicitly highlighted his government's plan to harness DPI to drive growth and inclusion, placing South Africa firmly within the global DPI expansion narrative. The country's position is significant: as the G20 host in 2025 and a leading voice in African digital governance, South Africa's DPI trajectory will influence how the continent's broader digital infrastructure agenda unfolds over the coming decade.

This creates both opportunity and responsibility. The decisions made now about security architecture, governance structures, and assurance mechanisms will shape not only South Africa's digital future but also serve as a reference model for the continent.

4.2 Risks and Implications: When Scale Becomes Vulnerability

The Blast Radius Problem

DPI systems differ fundamentally from traditional enterprise systems. Their blast radius is population-scale. A breach in an enterprise system affects an organization. A breach in a national identity system affects every citizen. A compromise of a payment rail affects an entire economy. These are not incremental differences in magnitude; they are qualitative differences in kind.

And certain compromises are irreversible, depending on how the initial system was set up. At times, biometric data cannot be reissued - A fingerprint, once exposed, is exposed permanently. An iris scan cannot be changed the way a password can. When DPI systems that hold biometric data are compromised, the damage extends across the lifetime of every affected citizen.

This was made viscerally clear in India. A data breach involving public exposure of personal data belonging to government officials affected more than 20 million citizens, illustrating that even the world's most sophisticated DPI deployment is not immune to systemic failure.

The WEF's Four Threat Categories

The World Economic Forum's October 2025 assessment of DPI security risks identified four categories of emerging threats that deserve explicit attention from any government deploying these systems:

1. **Synthetic identity fraud:** The combination of real and fabricated information to bypass traditional verification systems. With digital identities serving as the backbone for government services, banking, and telecommunications, cascading consequences are possible at the national scale.
2. **Algorithmic bias:** As DPI systems increasingly rely on AI for fraud detection and service delivery, ensuring that algorithms do not perpetuate inequalities is crucial, particularly in diverse populations with multiple languages, economic strata, and digital literacy levels.
3. **AI-powered cyber threats:** State-sponsored actors and criminal organizations are leveraging AI for targeted, adaptive attacks on critical infrastructure. For systems serving hundreds of millions of users, the potential for cascading failure demands entirely new cybersecurity approaches.
4. **Data sovereignty erosion:** Country-level risk falling into a "data trap" if critical national data is effectively controlled abroad. Local hosting does not always confer control; extraterritorial

laws can lead to “sovereignty washing,” in which services appear national but are governed from elsewhere.

Open Source: Shared Strength, Shared Exposure

The same mechanisms that enable DPI’s rapid deployment also create systemic risk. Shared components create shared exposure. A vulnerability in a widely deployed open-source component — say, a core element of MOSIP or a common authentication library — can affect multiple countries simultaneously. A single supply chain compromise can cascade across an entire ecosystem of deployments.

This was not a theoretical risk in 2025. Attacks on widely used open-source libraries, through dependency hijacking and repo jacking, affected thousands of downstream projects. For a DPG ecosystem deployed across 20+ countries, this attack vector carries strategic implications that go beyond any single organization’s threat model.

Cybersecurity in the DPG ecosystem is not a national issue. It is an ecosystem-level consideration that requires ecosystem-level responses.

The Capacity Gap in the Global South

Perhaps the most sobering statistic in the DPI security landscape: only 20% of low-income countries had a fully functional Computer Security Incident Response Team (CSIRT) in 2025. These are the digital equivalents of firefighters, and four out of five countries that deploy critical digital infrastructure lack them.

Meanwhile, cybercrime losses in India alone approached what analysts described as a significant national economic burden, with weekly cyberattacks averaging 3,195 per organization, 62% higher than the global average, driven in part by AI-enabled attack automation.

The math is unforgiving: rapid DPI expansion, combined with inadequate security capacity, creates a widening vulnerability gap that adversaries are actively exploiting.

External Collaboration: Opportunity and Dependency

International collaboration has been instrumental in advancing DPI. Estonia, Germany, the United States, and multilateral bodies have contributed technical expertise, proven frameworks, and capacity-building support. This is genuinely valuable. It has accelerated adoption and improved system design.

But as DPI becomes more central to national systems, the nature of these dependencies warrants careful examination. Consider the framing offered by one Harvard analysis of India’s DPI export strategy: “DPI is not a neutral toolset. It encodes assumptions about how the state should function, how identity is constructed, and what counts as legitimate data.”

This observation applies equally to DPI systems imported or adopted from any source. Technology transfers carry governance assumptions. Dependency structures carry leverage implications. For countries in the Global South seeking to build truly sovereign digital infrastructure, the question of long-term ownership of systems, of data, of decision-making is inseparable from the question of national security.

4.3 Pathways Forward: Building Security Into the Foundation

1. Security by Design, Not as an Afterthought

The most important shift in DPI security thinking is architectural: from treating security as a layer added on top of functioning systems, to embedding it as a core design principle from the outset. The Global DPI Summit in Cape Town in late 2025 crystallized this consensus: bold DPI rollouts must be paired with equally ambitious investments in digital resilience.

This means incorporating threat modeling into system design phases, not just deployment reviews. It means building resilience and recovery into architecture from day one. It means recognizing that security, safety, and public trust form a continuous chain in which each link depends on the others.

Most critically, most DPI incidents do not require sophisticated hacking. They exploit well-known, common vulnerabilities. Regular security updates, asset inventories, and consistently applied multi-factor authentication fundamentals would prevent a significant proportion of successful attacks. Simple does not mean easy. Consistent does not mean cheap. But the gap between available and deployed security practices remains shockingly wide.

2. Bug Bounty Programs: A Proven Early Signal

The bug bounty model has demonstrated concrete value in DPI contexts. India launched a bug bounty for Aadhaar in 2022. France followed with a co-designed program for its digital ID ecosystem via YesWeHack. In July 2025, M-Pesa, the dominant mobile money platform across multiple African markets, launched its own program in cooperation with HackerOne.

These programs have uncovered hundreds of critical vulnerabilities across DPI platforms, enabling mitigation before they are exploited. The challenge now is to scale this model across borders, including into lower-income contexts where bug bounty infrastructure and participation culture are less developed. International development partners have a concrete role to play here.

3. Strengthen and Fund CSIRTs

Computer Security Incident Response Teams are not optional infrastructure for countries deploying national DPI. They are as essential as the systems themselves. Yet only 20% of low-income countries had a fully functional CSIRT in 2025.

Closing this gap requires targeted multilateral investment, not merely technical assistance, but sustained capacity-building that builds institutional capability, retains trained personnel, and creates regional coordination mechanisms. The African Union, the G20 Digital Economy Working Group, and bilateral development partnerships all have roles to play in this effort.

4. Independent and Ecosystem-Level Assurance

As Digital Public Infrastructure (DPI) systems scale, assurance mechanisms must evolve beyond implementation-level assessments to include independent, ecosystem-wide perspectives. Existing efforts provide a strong foundation, but their impact can be significantly strengthened through complementary evaluations that bring cross-country visibility and multi-sector expertise.

Proposals from stakeholders such as DIAL and the G20 Digital Economy Working Group point to a coordinated global approach, supported by regional structures and shared standards, similar in spirit to frameworks developed by the International Civil Aviation Organization, which enhance interoperability and security without limiting national flexibility.

Looking ahead, meaningful assurance will require open collaboration among public institutions, open-source communities, and private-sector experts with deep domain expertise. This combined approach ensures that security is not assessed in isolation but continuously strengthened through

diverse expertise and independent validation. Ultimately, DPI resilience depends on building assurance as a shared capability, one that enables systems not only to function but to remain trusted, secure, and sustainable at scale.

5. Refining the Balance Between Openness and Strategic Discretion

Openness has been a defining strength of the DPI ecosystem. It has enabled knowledge sharing, rapid innovation, and global collaboration. These are real goods that should be preserved.

But as systems become more critical, it becomes important to balance openness with strategic discretion. Publicly available technical information can, in aggregate, reveal system structures, reduce the cost of reconnaissance for adversaries, and enable more targeted exploitation.

This is not an argument for secrecy. It is an argument for intentionality: share only what enables collaboration and keep private what could introduce vulnerability that could be exploited by threat actors masquerading as a DPI collaborator. This balance will require ongoing calibration as DPI systems continue to scale.

6. Cross-Border Vulnerability Disclosure and Shared Response

The ecosystem nature of DPI means that vulnerability disclosure and incident response must increasingly function across borders. A vulnerability discovered in one country's implementation of MOSIP may exist in nineteen others. A supply chain compromise affecting a shared component affects the entire ecosystem simultaneously.

Developing shared vulnerability disclosure frameworks, coordinated incident response protocols, and cross-country threat intelligence sharing, with appropriate sovereignty protections, is not merely aspirational. It is operationally necessary. The technical architecture for this exists. The political will and institutional frameworks to sustain it do not yet fully exist. That is the gap to close.

7. Managing AI-Enabled Cyber Threat Escalation in Digital Public Infrastructure

The integration of artificial intelligence into both attacker and defender toolkits has introduced a structural imbalance that current DPI security models are not designed to manage. AI does not simply increase the volume of attacks; it changes their nature, enabling speed, adaptability, and autonomy at a scale that compresses response timelines beyond human capability. As DPI systems increasingly embed AI into identity verification, fraud detection, and service delivery, they inherit not only the benefits of intelligence but also its vulnerabilities.

This requires a fundamental shift in how AI is understood within DPI: not as a feature layer, but as a security-critical component that must be governed, tested, and continuously monitored. Attackers are already exploiting AI systems through adversarial inputs, data poisoning, and prompt manipulation, while simultaneously using AI to generate more sophisticated intrusion techniques. The result is a dual exposure, AI as both a target and an enabler of attack.

Closing this gap demands that AI risk be treated as integral to DPI security architecture. This includes embedding AI-specific threat modeling into system design, establishing governance frameworks that define accountability and oversight, and ensuring that AI-driven decisions can be audited, challenged, and reversed. Without this, DPI systems risk evolving into environments where autonomous systems interact under adversarial conditions with limited human visibility or control.

The challenge is not hypothetical. It is already unfolding. “The question is whether DPI systems will evolve to manage AI-driven threats as a core security concern or absorb these risks incrementally until failure occurs at scale.”

V. The Diplomatic Dimension: Sovereignty, Norms, and Strategic Positioning

5.1 DPI as Geopolitical Instrument

Digital Public Infrastructure has evolved into something neither its architects nor its critics fully anticipated: a strategic instrument of geopolitics. India's positioning of its DPI model as a global public good represents a form of digital non-alignment, offering countries an alternative pathway to build sovereign, interoperable, and inclusive systems without defaulting to dominant global technology paradigms. For countries across Africa, Asia, and Latin America, this presents a compelling third path between competing models of digital power, thereby creating a viable third path in an increasingly polarized digital landscape:

- Reduces dependence on global platform ecosystems and centralized control models
- Enables interoperable, open, and scalable digital systems
- Supports public-interest governance aligned to national development priorities
- Accelerates digital inclusion and economic participation at the population scale

This shift elevates DPI from a technical enabler to a mechanism of strategic influence and alignment. Its repositioning of DPI within the geopolitical landscape is significant, as it shifts digital infrastructure from a technical domain into a space of strategic influence and alignment.

However, digital diplomacy is not neutral, and DPI adoption is not without consequence.

Technology transfer carries embedded governance assumptions, design philosophies, and long-term dependency implications that extend beyond immediate functionality. The open-source nature of DPI components does not inherently guarantee sovereignty, particularly where institutional capacity, technical expertise, and regulatory control remain uneven. In practice, this introduces a set of structural risks that must be understood and actively managed:

- Governance models embedded in imported systems may shape national policy outcomes
- Reliance on external expertise can create long-term operational dependencies
- Data sovereignty may be constrained by hosting, jurisdiction, and legal exposure
- Interoperability can introduce cross-border vulnerabilities if not securely governed
- Ownership and control of critical digital assets may become fragmented over time

DPI adoption must be treated not simply as a technical deployment but as a strategic national infrastructure decision, in which sovereignty, control, and resilience are as critical as deployment, and in which long-term implications outweigh short-term implementation gains.

5.2 The Norm Gap in Cyberspace

The most consequential diplomatic failure of the cyber era is the absence of binding international norms governing state conduct in cyberspace. Existing frameworks, such as the Tallinn Manual, the Budapest Convention, and the UN GGE process, have made important contributions. But they fall well short of the deterrence and accountability architecture needed for the current threat environment.

Consider what is currently permissible, or at least unpunished:

1. **State-sponsored actors can conduct sustained attacks**, with infrastructure mapping and pre-positioning operations in other nations' systems without triggering treaty obligations or meaningful diplomatic consequences.
2. **Proxy groups can be used to conduct offensive cyber operations** with deniability structures that are legally convenient even when technically transparent.
3. **Election interference through cyber** means has touched 18 countries in 2025 alone, with attribution frequently contested despite technical evidence.
4. **Critical infrastructure attacks on hospitals, energy grids, and payment systems** occur without the kind of international response that comparable kinetic attacks would trigger.

The cost of this norm gap is not abstract. It is measured in hospital patient mortality rates, supply chain disruptions, financial losses, and the gradual erosion of public trust in digital institutions.

5.3 A Diplomatic Agenda for the Next Phase

The emergence of DPI as contested national infrastructure extends security beyond technical domains into diplomacy, where coordination, norms, and collective accountability will determine whether risks are contained or amplified globally.

As systems become interconnected, no country can secure DPI in isolation; resilience is now defined at the ecosystem level, and vulnerability in one context can propagate across many.

Five diplomatic priorities are particularly actionable:

1. **Elevate DPI Security as a Global Policy Priority**

DPI security must become a standing agenda item in global diplomatic forums with clear outcomes on shared standards, governance models, and cross-border assurance. Without sustained political focus, efforts will remain fragmented and insufficient against systemic risks.

2. **Advance Binding Cyber Norms and Accountability**

Current voluntary frameworks are inadequate. Binding commitments regarding state conduct, particularly in critical infrastructure and cyber operations, are essential to closing the accountability gap and strengthening deterrence.

3. **Reframe Development Finance Around Security**

DPI security must be treated as a core requirement in development funding. Systems deployed without CSIRTs, security-by-design, and assurance mechanisms are not successes; they are deferred vulnerabilities with future national consequences.

4. **Enable Cross-Border Assurance and Coordinated Response**

Shared DPI components require shared security responses. Countries must establish coordinated vulnerability disclosure, threat intelligence sharing, and joint response mechanisms to effectively manage ecosystem-level risks.

5. **Align Digital Sovereignty with Collective Security**

Sovereignty must be balanced with interoperability and shared security baselines. Uncoordinated approaches risk fragmentation and uneven protection, weakening overall resilience across interconnected systems.

These priorities are foundational, not optional. In a globally connected threat environment, the security of any one system is only as strong as the ecosystem it operates within—and without coordinated diplomatic action, that weakest link will define the risk for all.

VI. Conclusion: The Foundation Beneath the Foundation

There is a principle in civil engineering worth importing into the digital domain: a structure's strength is determined not by its most visible elements but by its foundation. The grandest building stands or falls on what lies beneath the surface.

DPI systems have already demonstrated their value. They have enabled inclusion, reaching populations that formal financial and government systems had never reached before. They have improved service delivery at scale. They have driven economic growth and reduced transaction costs by orders of magnitude. These are real achievements, earned through institutional will, technical ingenuity, and patient investment.

The task ahead is not to question those achievements. It is to ensure they endure, and this requires facing a set of uncomfortable realities squarely:

- 1) **Cyberwar is not an external threat to DPI.** It is the operating environment in which DPI exists.
- 2) **The illusion of stability, systems functioning as normal** while adversaries probe, position, and accumulate access, is one of the most dangerous conditions a system can be in.
- 3) **AI has fundamentally altered the speed and scale of cyber operations**, and the advantage it currently provides to well-resourced attackers over resource-constrained defenders is a structural problem, not a temporary one.
- 4) **The DPI security gap in the Global South**, measured in missing CSIRTs, absent security-by-design standards, and insufficient assurance mechanisms, is not technical. It is a governance gap that requires political will to close.

“The success of this next phase will not be measured by how quickly systems are built, but by how well they endure under pressure.”

The convergence of cyberwarfare and Digital Public Infrastructure defines a new era in national security. As DPI systems become more central to national life and increasingly embedded in identity, payments, and access to government services, strengthening this foundation becomes a shared responsibility among governments, DPG communities, international partners, and ecosystem stakeholders.

The road ahead will be defined not by what has already been built, but by what remains structurally unaddressed. As DPI systems scale faster than the security, governance, and assurance mechanisms required to sustain them, the gap between adoption and resilience will continue to widen. This is not a distant or theoretical risk; it is an accumulating condition where systemic exposure grows quietly beneath the surface until it manifests at scale, often at the worst possible moment.

The challenge is immediate and structural: ensuring that the same systems designed to deliver inclusion and efficiency can withstand sustained pressure in an environment where stability can no longer be assumed and where failure carries consequences measured at a national, not technical, scale.

Digital infrastructure is no longer just a development tool. It is a strategic asset that must be sustained, protected, and trusted over time.

“The battlefield is invisible. The stakes are not.”

Strategic Recommendations Summary

The following recommendations represent a targeted yet comprehensive response to the structural, technical, and governance challenges outlined in this position paper. They are not theoretical constructs, but practical interventions designed to address the most critical gaps exposed by the convergence of cyberwarfare and Digital Public Infrastructure.

From the absence of threat-led design approaches and weak national cyber readiness to systemic risks created by shared digital components and the lack of independent assurance, each recommendation maps directly to a failure point identified in the preceding analysis. Collectively, they provide a layered pathway to strengthen resilience, improve security maturity, and align DPI deployment with its emerging role as critical national infrastructure.

1. Mandate Risk Assessment and Threat Modelling as the First Gate:

Require DPI-specific threat modeling (STRIDE/PASTA adapted for DPI) and risk assessments as mandatory preconditions before design, procurement, or deployment. No system should proceed without a clearly articulated threat model grounded in adversary capability and national context.

2. Adopt Security-by-Design as Mandatory DPI Standard:

Embed security-by-design principles at the software and infrastructure level, including resilience planning and security review, as non-negotiable phases of DPI design and procurement — not post-deployment additions.

3. Mandate Independent External Security Validation (Most Critical Control Before Go Live):

Require regular, independent, third-party security assessments across all DPI systems and DPG components, conducted by entities with no design, implementation, or operational conflict of interest. This must include ecosystem-level evaluation, not just country-level audits, to identify systemic risks that no single actor can see in isolation.

4. Fund and Mandate CSIRT Development:

International development partners must treat CSIRT capacity as a prerequisite for DPI funding, not an optional supplement. Target: full CSIRT functionality in all DPI-deploying low-income countries by 2028.

5. Scale Bug Bounty Infrastructure Across the DPG Ecosystem:

Develop multilateral mechanisms for cross-border bug bounty programs covering shared DPG components, with coordinated vulnerability disclosure and response protocols.

6. Establish an Independent Global DPI Assurance Body:

Create a global DPI assurance institution with regional chapters, standardized audit processes, private sector participation, and public reporting obligations, modeled on international standards bodies such as ICAO.

7. Advance Binding Cyber Norms in Multilateral Forums:

Use G20, AU, and UN GGE processes to move from voluntary principles toward binding commitments on state conduct in cyberspace, supported by accountability and attribution mechanisms.

8. Develop Cross-Border Vulnerability Disclosure Frameworks:

Establish coordinated vulnerability disclosure, shared threat intelligence, and incident response protocols across DPI-deploying nations, with appropriate sovereignty protections.

9. Reframe DPI Sovereignty as a Security Issue:

Require explicit clarity on data governance, ownership structures, infrastructure dependencies, and extraterritorial legal exposure as standard conditions of DPI deployment and technology transfer agreements.

10. Integrate AI Threat Assessment into DPI Security Planning:

Mandate AI-specific threat modelling across identity, payment, and data systems, addressing risks such as synthetic identity fraud, adversarial inputs, autonomous intrusion, and AI-enabled supply chain compromise.

11. Establish a DPI-Specific Cybersecurity Framework and Maturity Model:

Develop and adopt a standardized DPI cybersecurity framework with a measurable maturity model (L1–L5), enabling countries and funders to benchmark readiness, track progress, and condition investment on minimum security thresholds.

12. Strengthen DPG Supply Chain Security and SBOM Enforcement:

Require a Software Bill of Materials (SBOM), SLSA compliance, and continuous monitoring of dependency risk for all DPG components deployed in national infrastructure, with mandatory disclosure and patch SLAs.

13. Build DPI Security Operations and Continuous Monitoring Capability:

Establish dedicated DPI-focused Security Operations Centers (SOCs), integrating real-time monitoring, threat intelligence feeds, and incident response tailored to population-scale infrastructure.

14. Implement Zero-Trust Architecture and Identity-Centric Security Models:

Enforce zero-trust principles across DPI ecosystems, ensuring continuous authentication, least-privilege access, strong identity assurance, and secure API interaction across all components and participants.

15. Establish AI Governance and Security Frameworks for DPI:

Implement formal AI governance aligned to frameworks such as ISO/IEC 42001 and NIST AI RMF, including model registries, bias monitoring, adversarial testing, human oversight requirements, and AI incident response protocols.

While these measures are framed through the lens of national security and cyberwarfare, their relevance extends far beyond this context. They form the foundation of a broader DPI cybersecurity agenda that must evolve as the global expansion of digital public systems continues.

My next planned position paper builds on this work by examining DPI cybersecurity more holistically, expanding beyond the immediate pressures of cyber conflict to address the long-term governance, ecosystem coordination, and institutional design required to secure DPI as a sustainable global public good.

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About the Author

Godfrey Kutumela (He/Him)

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Cybersecurity Leader | DPI & DPG Security Architect | Digital Trust Advisor

Godfrey Kutumela is a globally recognized cybersecurity practitioner with over 26 years of experience shaping secure digital ecosystems across fintech, identity, and public infrastructure environments. He operates at the intersection of Digital Public Infrastructure (DPI), Digital Public Goods (DPGs), and national-scale platforms, bringing deep expertise in securing systems where the impact of failure is measured at the population level rather than the enterprise level.

- Trusted advisor to governments, foundations, and ecosystem builders delivering secure-by-design DPI across Africa, Latin America, and global markets
- Core contributor to large-scale platforms including Mowali (interoperability), Mojaloop (open payments), MOSIP (digital identity), and Tazama (fraud systems)
- Designs and embeds independent, risk-based security practices across payment systems, identity frameworks, and cross-border financial infrastructure
- Works across policy, architecture, and implementation layers, translating global standards into deployable security controls

His work spans both platform-level and ecosystem-level assurance, strengthening trust in open-source and government-led digital systems.

- Contributor to Digital Public Goods Alliance (DPGA) Privacy Expert Group, shaping global privacy and security indicators
- Active in open-source security, DevSecOps integration, and supply chain assurance for reusable digital platforms
- Delivered independent security reviews, penetration testing, and architecture governance across critical DPI implementations

Godfrey's leadership philosophy in the DPI ecosystem is clear: Security forms the foundation upon which safety and trust are built; safety being what citizens experience in practice, and trust is the legitimacy that sustains adoption, once lost, it is difficult to restore.

About Nucleus Systems

Nucleus Systems operates at the forefront of Digital Public Infrastructure (DPI) cybersecurity, shaping how security, trust, and resilience are defined for population-scale digital systems. Its work is grounded in the recognition that DPI has evolved into critical national infrastructure operating within a persistent cyber threat environment, requiring independent, ecosystem-level assurance beyond traditional approaches.

By combining deep technical expertise with alignment on governance, policy, and standards, Nucleus Systems addresses systemic risks across Digital Public Goods (DPG) and national deployments, positioning security as the foundation for safe, trusted, and sustainable digital transformation.

With a recent history of involvement in globally deployed platforms such as Mojaloop, Mowali, and Tazama, and contributions to global standard-setting through the Digital Public Goods Alliance, Nucleus Systems brings a rare practitioner-led, cross-ecosystem perspective.

This enables it to identify and address risks that are invisible at the individual system or country level, and to design scalable frameworks, maturity models, and independent assurance mechanisms.

As such, Nucleus Systems is not simply a service provider, but a strategic partner in advancing secure DPI adoption, offering funders and governments a high-leverage capability to strengthen resilience, reduce systemic risk, and safeguard the long-term legitimacy of digital public systems.